

Project 2 Planning Stage — transcript

- **Project:** ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator
- **Repository:** <https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator>
- **Branch:** project-2 (HEAD at start of session: dacb4aa; at save: dacb4aa, with spec.md and plan.md changed but not committed)
- **Date:** 2026-09-26
- **Learner:** RG Sureshkumar
- **Agent:** Claude Code (Opus 5.5)
- **Stage command:** start planning (instructions in plan.md, Commands section)
- **Outcome:** Implementation approach and six-checkpoint task sequence recorded in plan.md; two planning-stage revisions to spec.md (energy shown inside the new panels rather than as a third metric; complete removal of the output-only "agent" size); the learner approved the spec revisions and the plan and directed the agent to tick all three approval gates.

Note on redaction. Sanitised following transcripts/README.md: the author's email address, macOS home path and username, and temporary/session paths are replaced with placeholders. No messages were removed or reworded.

Note on tool calls. Each tool call is shown as a one-line summary in square brackets, followed by its output truncated to 400 characters. The agent's internal reasoning is not included. The complete local session log is kept locally and not committed. Automatic system notices shown to the agent during the session (such as file-change notices) are omitted.

Note on the final turn. The last learner turn (save transcript) is followed by the tool calls that generated this file; the agent's closing confirmation was written after the file was generated and so is not included.

1. Learner (RG Sureshkumar)

start planning

Agent

[Tool: Bash — Read plan instructions and list project files]

```
Exit code 1
# Implementation Plan
```

```
> EDITING DIRECTIVE: USER AND AGENT EDIT THIS FILE COLLABORATIVELY. THE USER MUST REVIEW AND APPROVE ITS CONTENT.
```

Purpose of this file: Turn the approved specification into ordered, updatable implementation and verification work.

Instructions for the user

Preserve the approved requirements and verify the completed work. Direct priorities, scope, and meaning ... [truncated]

[Tool: Bash — Read brief and specification]

<persisted-output>

Output too large (48.4KB). Full output saved to: <session-log>

Preview (first 2KB):

Project 2 Brief

> EDITING DIRECTIVE: DO NOT EDIT THIS FILE ... [truncated]

[Tool: Read — bq0bws0p4.txt]

1 # Project 2 Brief

2

3 > EDITING DIRECTIVE: DO NOT EDIT THIS FILE. IT IS THE AUTHORITATIVE PROJECT FRAME.

4

5 Roles in the project documents: The user is the person directing and working with the AI agent. The agent supports the user's work but does not make or approve decisions on the user's behalf. Employees are the intended audience for the completed calculator, and the named reference profi ... [truncated]

[Tool: Bash — Outline research and list code files]

1:# Research

7:## Instructions for the user

9:You are responsible for the ethics, accuracy, and fairness of the research. Direct the inquiry toward useful questions, judge sources and suggestions rather than accepting them at face value, and approve only results supported by verified evidence and audience needs. Seek evidence that challenges your assumptions, represent uncertainty honestly, and re ... [truncated]

[Tool: Bash — Check research approval status and selected features]

3:> EDITING DIRECTIVE: USER AND AGENT EDIT THIS FILE COLLABORATIVELY. THE USER MUST REVIEW AND APPROVE ITS CONTENT.

9:You are responsible for the ethics, accuracy, and fairness of the research. Direct the inquiry toward useful questions, judge sources and suggestions rather than accepting them at face value, and approve only results supported by verified evidence and audience needs. Seek evidence ... [truncated]

Both prerequisites are approved (features on 2026-09-26, spec on 2026-09-26). Now I'm reading the code so I can orient you before we plan.

[Tool: Bash — Read app, store, core, and data scripts]

```
<persisted-output>
Output too large (32.9KB). Full output saved to: <session-log>
```

Preview (first 2KB):

```
/**
 * app.js – boots the calculator and every company featur ... [truncated]
```

[Tool: Read — bl6di4v1w.txt]

```
1  /**
2   * app.js – boots the calculator and every company feature attached to it.
3   *
4   * Each module registers its renderer through AIPF.onUpdate during init, so the
5   * single emitUpdate() below paints the whole page once everything is wired.
6   */
7   (function (AIPF) {
8     'use strict';
9
10    function init() {
11      const root = document.getElementById('aipf');
12      if (!root || root.da ... [truncated]
```

[Tool: Bash — Inspect tracker and how features hook in]

```
/**
 * tracker.js – FEATURE 1: Live session tracker.
 *
 * The original calculator asks you to describe an average day. That is the
 * wrong question for someone who uses AI all day at work: nobody knows their
 * own average, and the estimate never changes as the day goes on. This logs
 * prompts as they actually happen, keeps a running total that updates live,
 * and projects where the day is hea ... [truncated]
```

[Tool: Bash — Check whether an energy view exists in the page]

```
assets/js/calculator.js:24:    p.set('k', state.metric); p.set('l', state.loc); p.set('h',
state.home);
assets/js/calculator.js:33:    const k = p.get('k'); if (k === 'carbon' || k === 'water')
state.metric = k;
assets/js/calculator.js:136:    const total = AIPF.aiDaily(state.metric);
assets/js/calculator.js:140:    const per =
AIPF.perPromptTriple(AIPF.getModel(row.model), row.size, state.met ... [truncated]
```

I've finished reading. Here's where things stand before we plan.

Prerequisites are met. You approved the five features in `research.md` and the specification in `spec.md`, both on 2026-09-26. One thing to know: sources S59–S80 are still marked "not yet assessed by the user". That doesn't block planning, but checking them should go on the verification list.

How the code works now: - The AI total is built from `state.rows`, each one a model, an output size and a count (`aiDailyTriple` in [core.js:238](#)). - The tracker stores each prompt as `{t, m, s}` and turns the

log back into those rows ([tracker.js:98](#)). - The budget reads each day's total from `trackerDayTotal`.

Images (feature 1) and agent sessions (feature 2) don't fit the model/size/count shape. So the first building task is a small foundation: tracker entries that can hold several kinds of item, and one AI-total function that the calculator, budget and tracker all share. The digital-day total (features 3 and 4) sits in a separate module and never feeds into that AI total.

Build order I'd suggest (it follows the dependencies): 1. Foundation for the AI total 2. Feature 1, images 3. Feature 2, agent sessions 4. Feature 3, digital day 5. Feature 4, device manufacturing 6. Feature 5, comparisons

Feature 5 goes last because its charts need both totals and its device items come from feature 4. Each step would end with the spec's acceptance checks run in the footprint preview.

One gap to settle first. The spec refers to an "energy view", for example feature 4 check 5: "In the energy view, manufacturing is not added". It also asks for results in energy, carbon and water. But the page only has a **Carbon / Water** switch. Energy appears only as a line of text, and none of the comparison items has an energy figure.

Which do you want? - **(a)** Add a third "Energy" option to the switch across the page. This is more work, and the comparison charts would have no everyday items to show in that view. - **(b)** Keep the Carbon / Water switch, and show energy as a figure with its range inside the new panels (images, sessions, digital day, devices). Feature 4 check 5 would then be reworded to "the energy figures don't include manufacturing, and a note says why".

Either way, the spec's wording has to match what gets built. Once you decide, I'll draft the matching spec wording for your review, record it under Revisions, and then draft the plan's Approach section.

2. Learner (RG Sureshkumar)

go with option b

Agent

[Tool: Bash — Find references to metric views in the spec]

```
15:Guide the specification one feature at a time. Help turn approved decisions into precise
requirements and surface gaps or trade-offs without inventing requirements or making
product decisions. Draft concise updates for review, focus on the intended result rather
than implementation steps, and never approve the specification on the user's behalf.
126:4. Sessions tagged with the same project name ... [truncated]
```

[Tool: Bash — Reword the spec's energy-view references and record the revision]

```
spec.md | 5 +++--
1 file changed, 3 insertions(+), 2 deletions(-)
```

[Tool: Bash — Find other uses of the agent size and page sections]

```
assets/js/team-data.js:39:      { model: 'claude-opus-4-8',      size: 'agent',  
count: 1.1 },  
assets/js/team-data.js:40:      { model: 'claude-sonnet-4-6',      size: 'agent',  
count: 2.4 },  
assets/js/team-data.js:49:      { model: 'claude-opus-4-8',      size: 'agent',  
count: 0.6 },  
assets/js/budget.js:19:  // never across providers: moving an agent session from Opus  
straight ... [truncated]
```

Now I'll draft the Approach section into plan.md.

[Tool: Edit — plan.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/plan.md has been updated successfully.

I've made both changes. Neither is approved until you review it.

1. Spec change for your review in [spec.md](#): - In feature 4, "the energy view says so" now reads "the digital-day energy figure says so". - Feature 4's acceptance check 5 now reads: "The digital-day energy figure does not include manufacturing, and a note beside it says no source reports manufacturing energy." - I added a Revisions entry recording your decision and the reason for it.

Nothing else in the spec changed. The spec was approved before this edit, so please confirm you accept the new wording.

2. Draft Approach in [plan.md](#), which covers: - **Structure**: new files for the digital day and devices, and the tracker extended for images and sessions. - **Your energy decision**. - **How the two totals are computed**: a single shared AI-total function, and a separate digital-day function that nothing else reads. - **Build order**: foundation, then features 1 through 5, with each step closing on its spec acceptance checks. - **Risks**: six are listed.

One risk needs your call before feature 2 is built. The spec removes the output-only "coding / agent session" size, but three other parts of the calculator still use it: - the fictional team profiles in [team-data.js:39](#) - tracker history already saved as "agent" entries - saved share links

If we simply remove the size, the team view breaks and old logs fall to zero or throw errors.

How do you want old uses handled? - **(a)** Remove it from every menu, but keep its figures internally so old history and the team profiles still calculate. They would be labelled as the old output-only method. - **(b)** Remove it completely. Old "agent" log entries are dropped or ignored, and the team profiles are rewritten using other sizes. Rewriting the profiles changes their figures, which the spec doesn't cover, so this would also need a spec revision. - **(c)** Something else you'd prefer.

3. Learner (RG Sureshkumar)

go with option b

Agent

[Tool: Bash — Read team profiles and share-link parsing]

```
/**
 * team-data.js — the company roster the team dashboard reads.
 *
 * △ THE FIGURES BELOW ARE ILLUSTRATIVE SAMPLE DATA, NOT MEASURED USAGE.
 * They describe a fictional company so the dashboard has something to show out
 * of the box. Replace them before drawing any real conclusion.
 *
 * To wire this to real usage, drop this file and call AIPF.setTeamData() with
 * the same shape once your own ... [truncated]
```

[Tool: Edit — plan.md]

[Tool: Bash — Check how the team view labels its sample data]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief — AI Footprint Calculator/plan.md has been updated successfully.
```

```
assets/js/team.js:228:    const warn = root.querySelector('#aipf-tm-sample');
assets/js/team.js:229:    if (warn) warn.hidden = !data.sample;
index.html:236:    <p class="aipf-tm-sample" id="aipf-tm-sample" hidden>
```

I've recorded option (b) under Risks in [plan.md](#). While checking it I found one more thing the plan now covers.

Share links store each output size by its position in a list ([calculator.js:44](#)). If "agent" is simply deleted, every old link that mentions the trilogy rewrite would load as the wrong size. So the plan has old links read with the old list positions and skip any old "agent" row. Old "agent" entries in the tracker history are ignored in every total, and nothing is deleted from storage.

Next decision: rewriting the team profiles. The team dashboard uses fictional sample data, labelled as sample in the page. Two teams include "agent" rows ([team-data.js:39](#)): - Engineering: Opus 1.1 a day and Sonnet 2.4 a day - Data Science: Opus 0.6 a day

A team profile is a list of model, size and count. The new agent sessions are token-based and live only in the tracker, so they can't go into that list as they stand.

How do you want these profiles rewritten? - **(a)** Delete the agent rows and add nothing in their place. This is the simplest option. Engineering's and Data Science's sample totals would fall a lot, and the comparison between teams would change. - **(b)** Replace each agent row with a text size that is already in the calculator, such as "A long document". The sample stays roughly heavy for those teams, but the figures become our own invention, which the spec would have to state. - **(c)** Let team profiles hold agent sessions as token counts using feature 2's method. This is the most faithful option, but it stretches the team view beyond what the spec requires, so it counts as a scope expansion.

Whichever you choose, I'll draft the spec revision for your review.

4. Learner (RG Sureshkumar)

go with option a

Agent

[Tool: Bash — Draft spec revision for full agent-size removal and update plan]

```
plan.md | 31 ++++++
spec.md | 10 ++++++----
2 files changed, 37 insertions(+), 4 deletions(-)
```

I've drafted the spec revision in [spec.md](#) for your review: - **Feature 2's behaviour** now says removal is complete: - old "agent" tracker entries are ignored in every total - Engineering's and Data Science's sample "agent" rows are deleted, with nothing in their place, so those teams' sample totals fall - old share links load correctly and skip any "agent" row - **Acceptance check 3** now covers the team data, old tracker entries, and old share links. - **Revisions** has a planning-stage entry explaining the change.

[plan.md](#) now records the decision too.

Two spec changes are waiting for your confirmation: the energy display and the agent-size removal. Both edit a spec you already approved, so please read the diff and say whether you accept them.

The next planning step is to replace the placeholder Implementation checklist with concrete tasks, one group for each build step: foundation, then features 1–5.

Before I draft them, how big should a checkpoint be? It decides where each commit falls and when you verify. - **(a)** One checkpoint per build step, six in all. You verify each feature's acceptance checks, then it's committed. - **(b)** Smaller checkpoints inside each feature, for example data and calculation first, then interface, then methodology text. There are more commits and more verification points.

5. Learner (RG Sureshkumar)

go with option a

Agent

[Tool: Bash — Show current checklist section of the plan]

```
## Checklist
```

```
Replace or expand the implementation placeholders below with tasks specific to the approved specification.
```

Approval gates

- [] User has reviewed, verified, and approved the research claims and selected features
- [] User has reviewed and approved the specification
- [] User has reviewed and approved the implementation approach and task sequence

Implementation

- [] Rep ... [truncated]

[Tool: Bash — Draft concrete implementation checklist in the plan]

(Bash completed with no output)

[Tool: Edit — plan.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/plan.md has been updated successfully. (note: the file had been modified on disk since you last read it – the edit applied cleanly, but the file contains other changes not in your context. Read it before edits that depend on surrounding content. ... [truncated])

I've replaced the placeholder Implementation checklist in [plan.md](#) with six checkpoints, one per build step:

1. **Foundation:** entry kinds in the tracker log, one shared AI-total function, and a check that every current figure on the page stays the same.
2. **Feature 1:** images.
3. **Feature 2:** agent sessions, the project view and estimate, and full removal of the "agent" size.
4. **Feature 3:** the typical-day panel, activity logging, router overlap, and the digital-day total.
5. **Feature 4:** "My devices", the manufacturing component, and the new device comparison figures.
6. **Feature 5:** the comparison items, both totals as bars, the driving setting, and the methodology rewrite.

Each checkpoint ends the same way: I run that feature's acceptance checks from the spec, you verify them, and I commit only when you ask. I also added two tasks that apply at every checkpoint: - re-run all earlier acceptance checks, to catch anything a later feature breaks - check the page at phone width and in both light and dark themes

I added one item under Verification: you still need to assess sources S59–S80, which the build relies on. All approval gates and verification items are unticked, because only you can tick them.

Still waiting on you: - Confirm the two spec revisions: the energy display, and removing the "agent" size completely. - Review the Approach and the checklist in `plan.md`.

Please read through the checkpoint tasks. Is anything missing, misplaced, or out of scope that you want changed before you consider approving the plan?

6. Learner (RG Sureshkumar)

I approve the spec revisions and plan.md is good

Agent

[Tool: Bash — Record the user's approval of the spec revisions]

```
plan.md | 98 ++++++-----  
spec.md | 12 +++++---  
2 files changed, 104 insertions(+), 6 deletions(-)
```

I've recorded your approval of the two spec revisions in [spec.md](#), under User approval, quoted as you wrote it.

"plan.md is good" is close to approval, but the plan's third approval gate asks for your explicit approval, and I can't infer it on your behalf. Two ways to finish this: - **Say you approve the plan**, for example "I approve the plan". I'll then record it in `plan.md` with the date and your words, and tick the third gate as your recorded approval. - **Tick the gates yourself** in [plan.md](#).

The first two gates, for the research and features and for the specification, are also unticked in `plan.md`. The approvals are already on record in `research.md` and `spec.md`, but ticking them is still yours to do. Tell me if you want me to tick them to reflect those recorded approvals, or do it yourself.

Once the plan is approved, this planning stage is finished. The planning transcript is a required deliverable, so please say `save transcript` when you're ready. Nothing will be committed unless you ask.

7. Learner (RG Sureshkumar)

I approve the plan, tick all three gates

Agent

[Tool: Bash — Tick the three approval gates at the user's direction]

```
### Approval gates
```

```
Ticked at the user's direction on 2026-09-26 ("I approve the plan, tick all three gates").  
The research features were approved on 2026-09-26 (`research.md`), the specification on  
2026-09-26 with its planning-stage revisions (`spec.md`), and this plan on 2026-09-26.
```

- [x] User has reviewed, verified, and approved the research claims and selected features
- [x] User has reviewed ... [truncated]

I've ticked all three approval gates in [plan.md](#) as you asked, with a note above them recording your approval in your words and dated 2026-09-26. The plan is approved, so the planning stage is complete.

Everything else in `plan.md` is still unticked: the implementation tasks, the verification items (including assessing S59–S80) and delivery.

This stage changed `spec.md` and `plan.md`, and nothing has been committed. I'll commit only when you ask.

The planning transcript is a required deliverable. Please say `save transcript` and I'll save this conversation to `transcripts/plan-YYYY-MM-DD_HHMMSS.md`.

When you're ready to build, open a fresh chat and type `start implementation`. It begins with checkpoint 1, the foundation for the AI total.

8. Learner (RG Sureshkumar)

`save transcript`

Agent

[Tool: Bash — Read transcript conventions, an example, and the PDF memory]

```
# Transcripts
```

```
Session transcripts for this project, kept as evidence of process.
```

- ``session-2026-09-12.md`` – the build session: porting Andy Masley's calculator to a standalone site and adding the three features.
- ``debrief-*.md`` – Unit 1 debrief conversations, saved separately so debrief evidence stays distinct from build sessions.
- ``debrief-*.pdf`` – the same debrief rendered for submissio ... [truncated]

[Tool: Bash — Locate and inspect this session's log structure]

```
total 32696
drwxr-xr-x@ 14 <user>  staff    448 Sep 26 19:29 .
drwxr-xr-x@  9 <user>  staff    288 Sep 11 19:47 ..
-rw-----@  1 <user>  staff 5426151 Sep 20 12:26 100c399a-421a-4db4-a70b-
9e7865d9c41a.jsonl
drwx-----@  4 <user>  staff    128 Sep 26 19:30 2ac37143-43c0-47d7-a8c8-353de4defd40
-rw-----@  1 <user>  staff  964718 Sep 26 19:41 2ac37143 ... [truncated]
```